

Module End Assignment-MYSQL

Project Title: E-Commerce Customer Churn Analysis

Aim

This project aims to delve into the dynamics of customer churn within an e-commerce domain, utilizing historical transactional data to uncover underlying patterns and drivers of churn.

Project Steps and Objectives:

Data Cleaning:

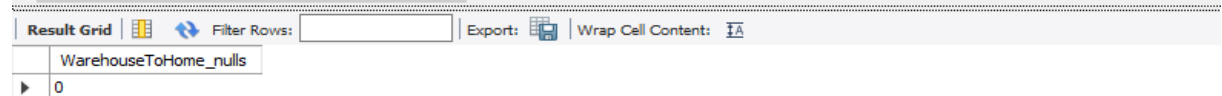
Handling Missing Values and Outliers:

- ❖ Impute mean for the following columns, and round off to the nearest integer if required: WarehouseToHome, HourSpendOnApp, OrderAmountHikeFromlastYear, DaySinceLastOrder.

Ans: Fig.1, Fig.2, Fig.3 & Fig.4 represents handling missing values by taking mean for the columns WarehouseToHome, HourSpendOnApp, OrderAmountHikeFromlastYear, DaySinceLastOrder.

Fig.1

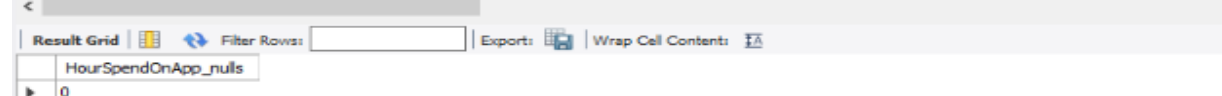
```
5663 -- WarehouseToHome
5664 • update customer_churn
5665 set WarehouseToHome = (select round(avg(WarehouseToHome)) from (select * from customer_churn) as t)
5666 where WarehouseToHome is null;
5667 • select
5668 sum(case when WarehouseToHome is null then 1 else 0 end) as WarehouseToHome_nulls from customer_churn;
```



WarehouseToHome_nulls
0

Fig.2

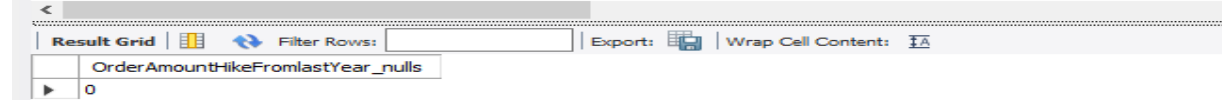
```
5674 -- HourSpendOnApp
5675 • update customer_churn
5676 set HourSpendOnApp= (select round(avg(HourSpendOnApp)) from (select * from customer_churn) as t)
5677 where HourSpendOnApp is null;
5678 • select
5679 sum(case when HourSpendOnApp is null then 1 else 0 end) as HourSpendOnApp_nulls from customer_churn;
```



HourSpendOnApp_nulls
0

Fig.3

```
5680
5681 -- OrderAmountHikeFromlastYear
5682 • update customer_churn
5683 set OrderAmountHikeFromlastYear= (select round(avg(OrderAmountHikeFromlastYear))
5684 from (select * from customer_churn) as t)
5685 where OrderAmountHikeFromlastYear is null;
5686 • select
5687 sum(case when OrderAmountHikeFromlastYear is null then 1 else 0 end)
5688 as OrderAmountHikeFromlastYear_nulls from customer_churn;
```



OrderAmountHikeFromlastYear_nulls
0

Fig.4

```

5690 -- DaySinceLastOrder
5691 • update customer_churn
5692   set DaySinceLastOrder= (select round(avg(DaySinceLastOrder))
5693   from (select * from customer_churn) as t)
5694   where DaySinceLastOrder is null;
5695 • select
5696   sum(case when DaySinceLastOrder is null then 1 else 0 end)
5697   as DaySinceLastOrder_nulls from customer_churn;

```

Result Grid

DaySinceLastOrder_nulls
0

❖ Impute mode for the following columns: Tenure, CouponUsed, OrderCount.

Ans: Fig.5, Fig.6, Fig.7 represents handling missing values by taking mode for the following columns Tenure, CouponUsed, OrderCount.

Fig.5

```

5700 -- Tenure
5701 • update customer_churn
5702   set Tenure = (
5703   select Tenure from (
5704   select Tenure from customer_churn
5705   group by Tenure
5706   order by count(*) desc limit 1) as t)
5707   where Tenure is null;
5708 • select
5709   sum(case when Tenure is null then 1 else 0 end)

```

Result Grid

Tenure_nulls
0

Fig.6

```

5712 -- CouponUsed
5713 • update customer_churn
5714   set CouponUsed = (
5715   select CouponUsed from (
5716   select CouponUsed from customer_churn
5717   group by CouponUsed
5718   order by count(*) desc limit 1) as t)
5719   where CouponUsed is null;
5720 • select
5721   sum(case when CouponUsed is null then 1 else 0 end)
5722   as CouponUsed_nulls from customer_churn;

```

Result Grid

CouponUsed_nulls
0

Fig.7

```
5724 -- OrderCount
5725 • update customer_churn
5726   set OrderCount = (
5727     select OrderCount from (
5728       select OrderCount from customer_churn
5729       group by OrderCount
5730       order by count(*) desc limit 1) as t)
5731   where OrderCount is null;
5732 • select
5733   sum(case when OrderCount is null then 1 else 0 end)
5734   as OrderCount_nulls from customer_churn;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Cor

OrderCount_nulls
0

- ❖ Handle outliers in the 'WarehouseToHome' column by deleting rows where the values are greater than 100

Ans: Fig.8 represents handling the outliers in 'WarehouseToHome' where the values are greater than 100.

Fig.8

```
5738 • delete from customer_churn
5739   where WarehouseToHome > 100;
5740 • set sql_safe_updates = 1;
5741 • select max(WarehouseToHome) as max_value from customer_churn;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

max_value
36

Dealing with Inconsistencies

- ❖ Replace occurrences of “Phone” in the 'PreferredLoginDevice' column and “Mobile” in the 'PreferredOrderCat' column with “Mobile Phone” to ensure uniformity.

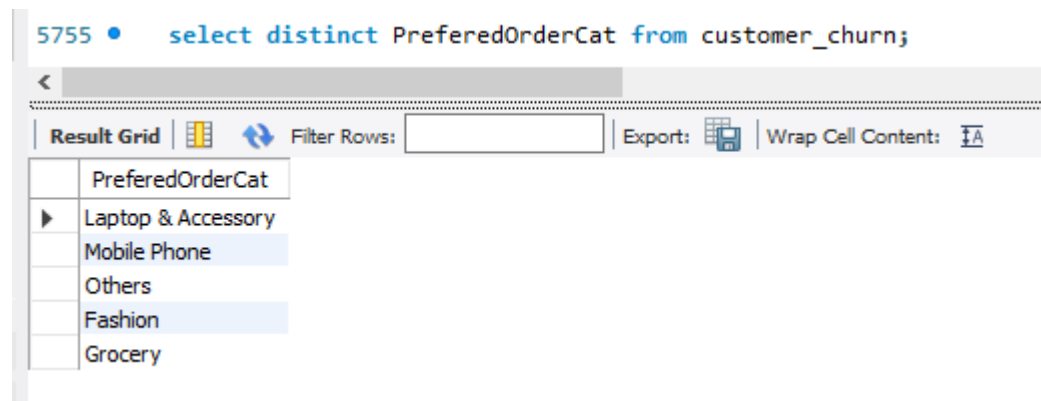
Fig.9: - Represents the replacing of “Phone” to “Mobile Phone” in column 'PreferredLoginDevice'.

```
5753 • select distinct PreferredLoginDevice from customer_churn;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

PreferredLoginDevice
Mobile Phone
Computer

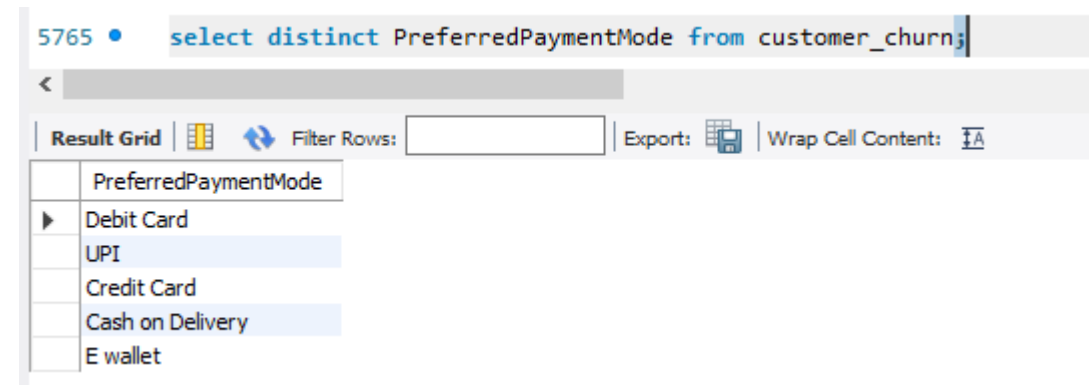
Fig.10: - Represents the replacing of “Mobile” to “Mobile Phone” in column 'PreferredOrderCat'.



- ❖ Standardize payment mode values: Replace "COD" with "Cash on Delivery" and "CC" with "Credit Card" in the PreferredPaymentMode column.

Ans: Fig.11: - Represents the replacing of values "COD" with "Cash on Delivery" and "CC" with "Credit Card" in the PreferredPaymentMode column.

Fig.11



Data Transformation:

Column Renaming:

- ❖ Rename the column "PreferredOrderCat" to "PreferredOrderCat".
- ❖ Rename the column "HourSpendOnApp" to "HoursSpentOnApp".

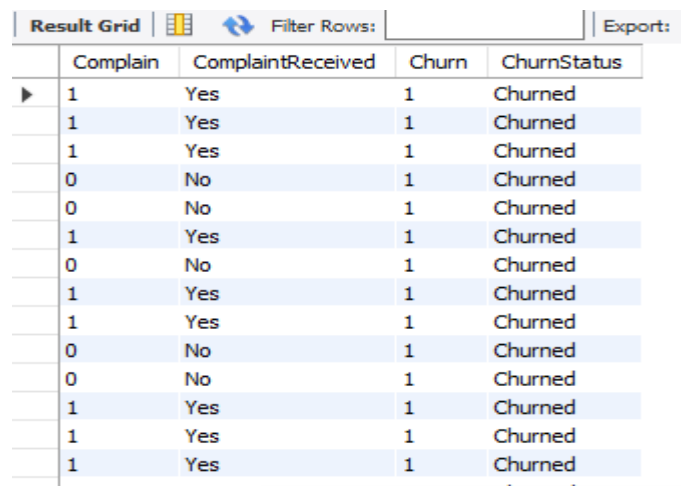
Ans: - Fig.12 represents the renaming of "PreferredOrderCat" to "PreferredOrderCat" and "HourSpendOnApp" to "HoursSpentOnApp".

HoursSpentOnApp	NumberOfDeviceRegistered	PreferredOrderCat
3	3	Laptop & Accessory
3	4	Mobile Phone
2	4	Mobile Phone
2	4	Laptop & Accessory
3	3	Mobile Phone
3	5	Mobile Phone
2	3	Laptop & Accessory

Creating New Columns:

- ❖ Create a new column named 'ComplaintReceived' with values "Yes" if the corresponding value in the 'Complain' is 1, and "No" otherwise.
- ❖ Create a new column named 'ChurnStatus'. Set its value to "Churned" if the corresponding value in the 'Churn' column is 1, else assign "Active".

Ans: - Fig.13 represents the creation of new columns named 'ComplaintReceived' and 'ChurnStatus' based on the corresponding values.



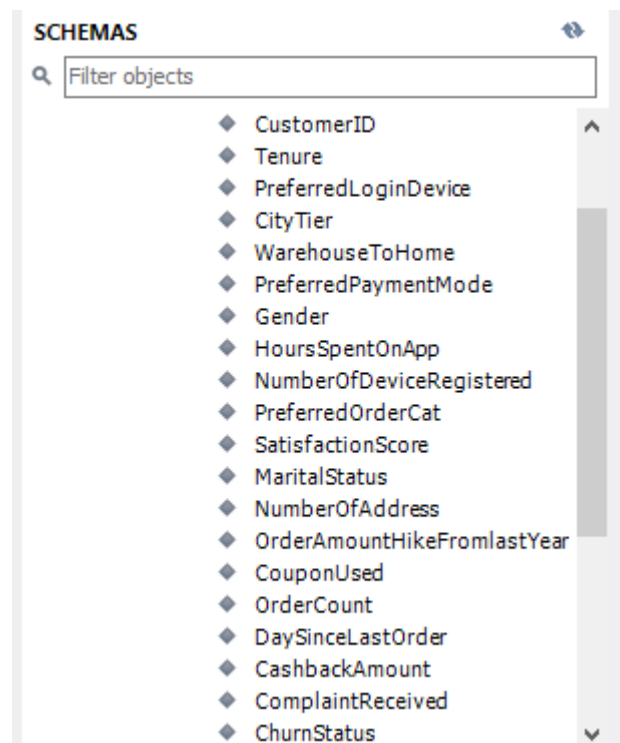
The screenshot shows a 'Result Grid' interface with a table containing four columns: 'Complain', 'ComplaintReceived', 'Churn', and 'ChurnStatus'. The 'Complain' column has values 1 or 0. The 'ComplaintReceived' column has values 'Yes' or 'No'. The 'Churn' column has values 1 or 0. The 'ChurnStatus' column has values 'Churned' or 'Active'. The table contains 15 rows of data.

	Complain	ComplaintReceived	Churn	ChurnStatus
▶	1	Yes	1	Churned
	1	Yes	1	Churned
	1	Yes	1	Churned
	0	No	1	Churned
	0	No	1	Churned
	1	Yes	1	Churned
	0	No	1	Churned
	1	Yes	1	Churned
	1	Yes	1	Churned
	0	No	1	Churned
	0	No	1	Churned
	1	Yes	1	Churned
	1	Yes	1	Churned
	1	Yes	1	Churned

Column Dropping:

- ❖ Drop the columns "Churn" and "Complain" from the table.

Fig.14: - Represents the dropping of columns "Churn" and "Complain".

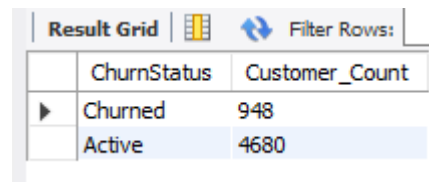


Data Exploration and Analysis:

- ❖ Retrieve the count of churned and active customers from the dataset.

Ans: The query groups the data by ChurnStatus and counts the customers, showing 948 Churned and 4680 Active customers.

Fig.15



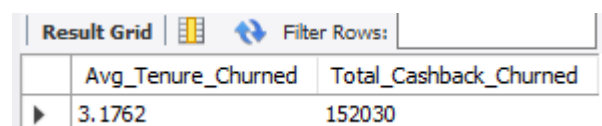
The screenshot shows a 'Result Grid' with a 'Filter Rows' button. The grid contains two columns: 'ChurnStatus' and 'Customer_Count'. There are two rows: 'Churned' with a count of 948, and 'Active' with a count of 4680.

ChurnStatus	Customer_Count
Churned	948
Active	4680

- ❖ Display the average tenure and total cashback amount of customers who churned.

Ans: - The query filters for churned customers and calculates their average tenure as 3.17 and total cashback as 152,030.

Fig.16



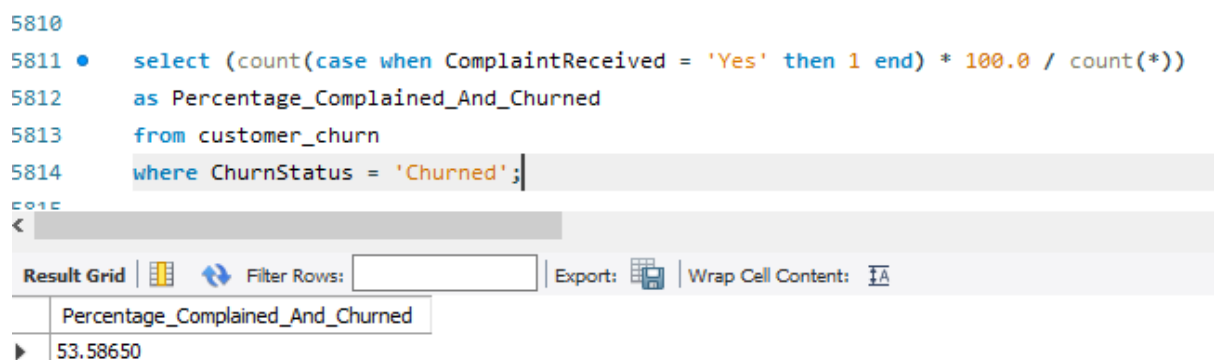
The screenshot shows a 'Result Grid' with a 'Filter Rows' button. The grid contains two columns: 'Avg_Tenure_Churned' and 'Total_Cashback_Churned'. There is one row showing an average tenure of 3.1762 and a total cashback of 152030.

Avg_Tenure_Churned	Total_Cashback_Churned
3.1762	152030

- ❖ Determine the percentage of churned customers who complained.

Ans: -The query calculates the share of churned customers who complained by counting complaints among churned users, showing 53.58% of churned customers had complained.

Fig.17



The screenshot shows a SQL query in a code editor and its result grid below it. The query calculates the percentage of churned customers who complained. The result grid shows a single row with the value 53.58650.

```
5810
5811 • select (count(case when ComplaintReceived = 'Yes' then 1 end) * 100.0 / count(*))
5812 as Percentage_Complained_And_Churned
5813 from customer_churn
5814 where ChurnStatus = 'Churned';
```

Percentage_Complained_And_Churned
53.58650

- ❖ Identify the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessory.

Ans: - The query filters churned customers with PreferredOrderCat = 'Laptop & Accessory', groups by CityTier and counts them, showing CityTier 3 has the highest churn for this category with 150 customers.

Fig.18

Result Grid			Filter Rows
	CityTier	Churned_Count	
▶	3	150	

- ❖ Identify the most preferred payment mode among active customers.

Ans: - The query groups active customers by PreferredPaymentMode and counts them, showing Debit Card is the most preferred payment mode with 1956 active customers.

Fig.19

	PreferredPaymentMode	Active_Customers
▶	Debit Card	1956

- ❖ Calculate the total order amount hike from last year for customers who are single and prefer mobile phones for ordering.

Ans: - The query filters for MaritalStatus = 'Single' and PreferredOrderCat = 'Mobile Phone' and sums the OrderAmountHikeFromLastYear, showing a total hike of 12,177 for this group.

Fig.20

	Total_Hike
▶	12177

- ❖ Find the average number of devices registered among customers who used UPI as their preferred payment mode.

Ans:

The query filters for PreferredPaymentMode = 'UPI' and calculates AVG(NumberOfDeviceRegistered), showing UPI users have on average 3.71 devices registered.

Fig.21

	Avg_UPI_Customers
▶	3.7174

- ❖ Determine the city tier with the highest number of customers.

Ans: - The query groups customers by CityTier and counts them, then orders descending to find the max, showing CityTier 1 has the highest number of customers with 3666.

Fig.22

	CityTier	Total_Customers
▶	1	3666

- ❖ Identify the gender that utilized the highest number of coupons.

Ans: - The query groups by Gender and sums CouponUsed, ordering by total descending, showing Male customers used the highest number of coupons with 5629.

Fig.23

	Gender	Total_Coupons
▶	Male	5629

- ❖ List the number of customers and the maximum hours spent on the app in each preferred order category.

Ans: - The query groups by PreferredOrderCat and calculates COUNT(*) as Num_Customers and MAX(HourSpentOnApp) as Max_HrsSpent, showing Mobile Phone has the most customers (2078) and the max time spent is 5 hours for Laptop, Mobile and Fashion, and 4 hours for Others and Grocery.

Fig.24

	PreferredOrderCat	Num_Customers	Max_HrsSpent
▶	Laptop & Accessory	2050	5
	Mobile Phone	2078	5
	Others	264	4
	Fashion	826	5
	Grocery	410	4

- ❖ Calculate the total order count for customers who prefer using credit cards and have the maximum satisfaction score.

Ans: - The query filters for PreferredPaymentMode = 'Credit Card' and SatisfactionScore = (MAX score - 5) and then does SUM(OrderCount), giving a total of 1122 orders for this group.

Fig.25

	Total_OrderCount
▶	1122

- ❖ What is the average satisfaction score of customers who have complained?

Ans: - The query filters for ComplainReceived = 'Yes' and calculates AVG(SatisfactionScore), showing the average satisfaction score of complained customers is 2.99, indicating near-neutral satisfaction.

Fig.26

	Avg_satisfaction_Complained
▶	2.9988

- ❖ List the preferred order category among customers who used more than 5 coupons.

Ans: - Result shows even heavy coupon users prefer Laptop & Accessory: 99 – highest, Fashion: 89, Mobile Phone: 45, Grocery: 42, Others: 28

Fig.27

	PreferredOrderCat	Morethan5_CouponUsed
►	Laptop & Accessory	99
	Fashion	89
	Mobile Phone	45
	Grocery	42
	Others	28

- ❖ List the top 3 preferred order categories with the highest average cashback amount.

Ans: - The query groups by PreferredOrderCat and calculates AVG(CashbackAmount), then orders descending with LIMIT 3. Others: 304.45 - highest avg cashback.

Fig.28

	PreferredOrderCat	Highest_avgCashback
►	Others	304.4545
	Grocery	266.2366
	Fashion	210.4031

- ❖ Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.

Ans: - Result is empty, it means no payment mode satisfies both conditions together. There is no group where average tenure is exactly 10 months and total orders are > 500.

Fig.29

	PreferredPaymentMode	Avg_Tenure
--	----------------------	------------

- ❖ Categorize customers based on their distance from the warehouse to home such as 'Very Close Distance' for distances <=5km, 'Close Distance' for <=10km, 'Moderate Distance' for <=15km, and 'Far Distance' for >15km. Then, display the churn status breakdown for each distance category.

Ans: - The Fig.30 shows Very Close Distance: 7 Active, 1 Churned (8 total), Close Distance: 1696 Active, 265 Churned (1961 total), Moderate Distance: 1106 Active, 184 Churned (1290 total), Far Distance: 1871 Active, 498 Churned (2369 total) - Highest churn count.

Fig.30

	Distance_Category	ChurnStatus	Customer_Count
►	Close Distance	Active	1696
	Close Distance	Churned	265
	Far Distance	Active	1871
	Far Distance	Churned	498
	Moderate Distance	Active	1106
	Moderate Distance	Churned	184
	Very Close Distance	Active	7
	Very Close Distance	Churned	1

- ❖ List the customer's order details who are married, live in City Tier-1, and their order counts are more than the average number of orders placed by all customers.

Ans: - Fig.31 shows list of such customers. Like Customer 50049, 50119, 50132 etc. They are all married, from Tier-1 city, and they ordered more than average.

Fig.31

```

5921
5922 • select * from customer_churn
5923     where MaritalStatus = 'Married'
5924     and CityTier = 1
5925     and OrderCount > (select avg(OrderCount) from customer_churn);

```

CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDeviceReg
50049	3	Computer	1	15	Credit Card	Male	3	3
50119	13	Mobile Phone	1	30	Cash on Delivery	Male	3	3
50132	15	Mobile Phone	1	16	Credit Card	Female	2	4
50133	13	Mobile Phone	1	8	Credit Card	Male	2	3
50152	2	Mobile Phone	1	28	Debit Card	Female	3	2
50180	12	Mobile Phone	1	6	Credit Card	Male	2	2
50187	18	Computer	1	13	Debit Card	Male	3	3
50229	10	Mobile Phone	1	12	Credit Card	Male	2	2

- ❖ Create a 'customer_returns' table in the 'ecomm' database and inserting values.

Ans: - Fig.32 shows the creation of table customer_returns in the ecomm database and inserted values.

ReturnID	CustomerID	ReturnDate	RefundAmount
1001	50022	2023-01-01	2130
1002	50316	2023-01-23	2000
1003	51099	2023-02-14	2290
1004	52321	2023-03-08	2510
1005	52928	2023-03-20	3000
1006	53749	2023-04-17	1740
1007	54206	2023-04-21	3250
1008	54838	2023-04-30	1990
NULL	NULL	NULL	NULL

- ❖ Display the return details along with the customer details of those who have churned and have made complaints.

Ans: - Only 3 customers match this. Like 50316, 52321, 53749. They left the company and they also made a complaint.

Fig.33

```

5949 • select c.*, r.ReturnID, r.ReturnDate, r.RefundAmount
5950     from customer_churn c
5951     join customer_returns r on c.CustomerID = r.CustomerID
5952     where c.ChurnStatus = 'Churned'
5953     and c.ComplaintReceived = 'Yes';

```

CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDeviceReg
50316	0	Computer	2	29	UPI	Female	2	3
52321	18	Mobile Phone	3	19	E wallet	Male	2	1
53749	1	Mobile Phone	3	31	Credit Card	Female	3	5

Conclusion

This project analyses customer churn to understand why customers leave. The results show that no payment mode has an average tenure of 10 months with over 500 orders, distance has a clear impact as customers living far from the warehouse (>15km) show the highest churn, while married customers from Tier-1 cities who order more than average are mostly active and valuable, and finally customers who filed complaints and made returns are highly likely to churn. Therefore, to reduce churn the company should improve delivery service for far-distance customers and resolve complaints faster to retain valuable customers.

Thank You ...